

Análise de Dados para Business Intelligence

Avaliação Teórico Prática 1

129055 – Vaag Mkrtchyan

1. Exploração inicial básica.

- a. The dimension of the dataset is **7000**.

'data.frame': 7000 obs. of 27 variables:

- b. Types of variables: **int, num, chr**

- c. **Recency**: 23 missing data

MntDrinks: 28 missing data

Recency	
Min.	: 0.00
1st Qu.	: 24.00
Median	: 49.00
Mean	: 49.24
3rd Qu.	: 74.00
Max.	: 99.00
NA's	: 23

MntDrinks	
Min.	: 0.0
1st Qu.	: 40.0
Median	: 180.0
Mean	: 545.7
3rd Qu.	: 700.0
Max.	: 3980.0
NA's	: 28

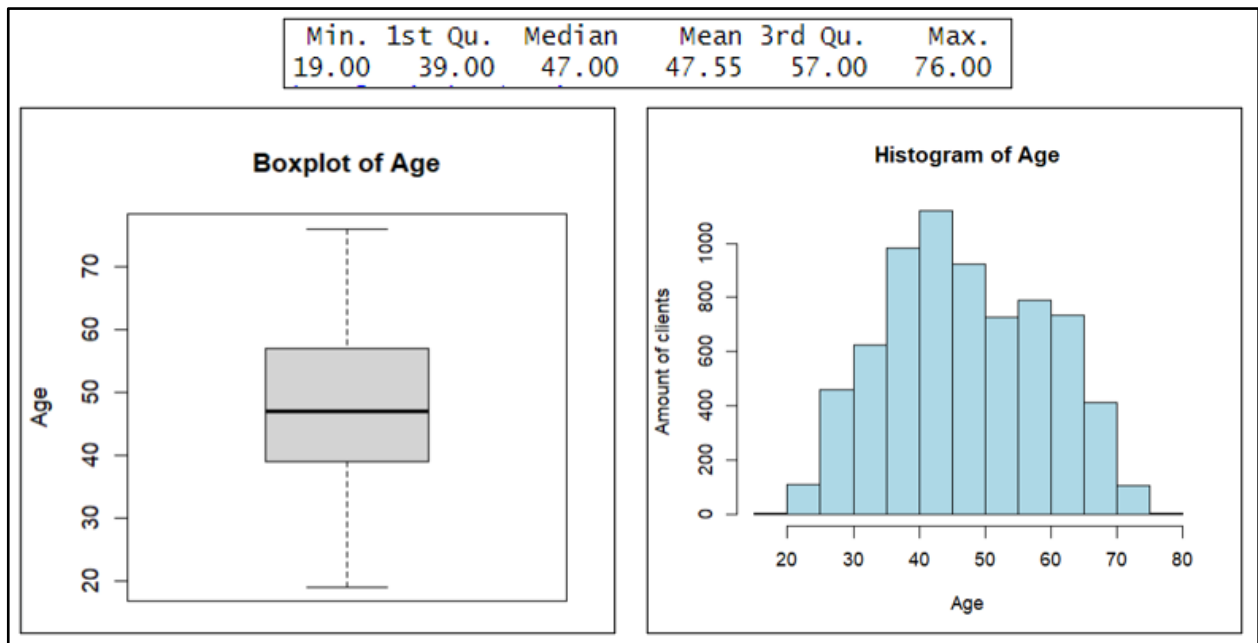
Also, there are 14 missing data in **Education**.

```
> blank_rows <- sabor[sabor$Education == "" | is.na(sabor$Education), ]  
> count_blank_rows <- nrow(blank_rows)  
> print(count_blank_rows)  
[1] 14
```

2. Construa novas variáveis.

- a. Statistics indicate that the variable has a range from 19 to 76, with the majority of values concentrated between the first and third quartiles, 39 and 57, respectively. The mean value of 47.55 is close to the median of 47, suggesting a symmetrical distribution around the center.

Also there are no outliers.



- b.

```
> str(sabor$TotalChildren)
int [1:7000] 1 1 1 1 2 1 0 1 2 1 ...
```

- c. I think it is reasonable to create a new variable that will indicate the total amount of money spent in the restaurant, by summarizing spendings in every category.

```
> str(sabor$MntTotal)
num [1:7000] 14582 9743 9968 8503 577 ...
```

3. Análise exploratória univariada do perfil demográfico

- a. *Marital Status*: ~65% of our clients are married, whereas only ~22% are single. That can be important for restaurant strategy.

Divorced	Married	Single	Widow
0.09214286	0.65314286	0.21971429	0.03500000

Children: More than half of our clients has more than one child. It can be also used for strategic development.

0	1	2	3	4
0.296571429	0.498714286	0.182857143	0.020714286	0.001142857

Education: it has many of unprepared data, so it's hard to analyze it, but I can see that most of clients (~87%) have a higher education.

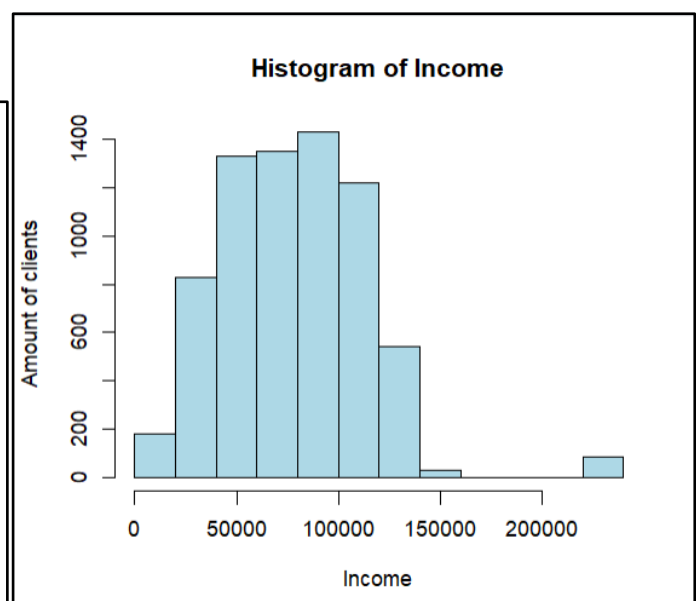
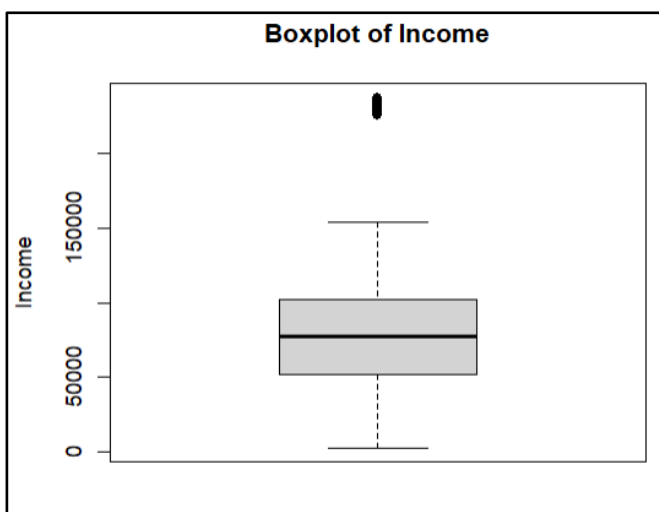
Graduation	highschool	HighSchool	master	Master	phd	PhD
0.4995714286	0.0002857143	0.0947142857	0.0010000000	0.1621428571	0.0002857143	0.2134285714

Income: The mean value of 47.55 is close to the median of 47, suggesting a relatively symmetrical distribution around the center.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
2494	51586	77190	77989	102016	237640

- b. Although the mean value of income is close to the median, by looking at charts I can see that there is a group of outliers, that are separated from the main part of clients.

I can assume that only income and age data is relevant as demographic numerical data. Age does not have any outliers.



- c. Education obviously contains **many unprepared data**. There are 14 **missing values**, also other values have **duplicates**, where the name starts with a small letter, rather than big. I suggest **deleting** irrelevant data, or **transforming** it into an appropriate format.

Also, I don't think there is a strategical need for a restaurant in separating people between higher education degree categories, such as bachelors, masters and PhD, because I don't think it is important. I suggest leaving only 'basic', 'higher school' and 'university graduation' categories, for more generic analyze.

4. Responda às seguintes questões

a. Entries: 534.7

Meat & Fish: 3080 (**highest**)

Vegan & Vegetarian: 2785

Drinks: 545.7

Desserts: 540.7

Additional Requests: 42.56

People spend more money on fish and meat.

As I can see, in every category

of food, there is a significant difference between mean and median. **Mean is times much higher than median.** It happens because high values “pull” the mean upward. Such high values can be **outliers**.

```
> summary(sabor$MntAdditionalRequests)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.00   9.00   24.00  42.56  57.00  249.00
> summary(sabor$MntDesserts)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0   40.0   180.0  540.7  680.0  3980.0
> summary(sabor$MntDrinks)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0   40.0   180.0  545.7  700.0  3980.0
> summary(sabor$MntEntries)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.0   40.0   180.0  534.7  680.0  3980.0
> summary(sabor$MntMeat.Fish)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0     250   1820   3080   5070  14980
> summary(sabor$MntVegan.Vegetarian)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0     240   1110   2785   3795  25974
> |
```

b. **Offer purchases:** 2.448

Application purchases: 6.016

Take away purchases: 3.835

Store purchases: 5.791

```
> summary(sabor$NumOfferPurchases)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   1.000   2.000   2.448   3.000   16.000
> summary(sabor$NumAppPurchases)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   4.000   6.000   6.016   8.000   13.000
> summary(sabor$NumTakeAwayPurchases)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   1.000   3.000   3.835   5.000   24.000
> summary(sabor$NumStorePurchases)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   3.000   5.000   5.791   8.000   13.000
```

c. There are **no clients** who made their card more than 2.5 years ago.

```
> str(subset(sabor, Time_Adherence > 30))
'data.frame':  0 obs. of  30 variables:
```

But there are 216 of them who made it exactly 2.5 years ago.

```
> str(subset(sabor, Time_Adherence >= 30))
'data.frame':  216 obs. of  30 variables:
```

- d. Mean amount of application visits per month is 5.278

```
> summary(sabor$NumAppVisitsMonth)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000   3.000   5.000   5.278   7.000  20.000
```

- e. I can see that ~99% of clients did not have complaints, so I can make a conclusion about great level of service in a restaurant.

```
> prop.table(table(sabor$Complain))

      0      1
0.98971429 0.01028571
```

- f. The most effective campaign was campaign number 1, with a ~8% acceptance rate.

```
> prop.table(table(sabor$Response_Cmp1))

      0      1
0.92085714 0.07914286
> prop.table(table(sabor$Response_Cmp2))

      0      1
0.92671429 0.07328571
> prop.table(table(sabor$Response_Cmp3))

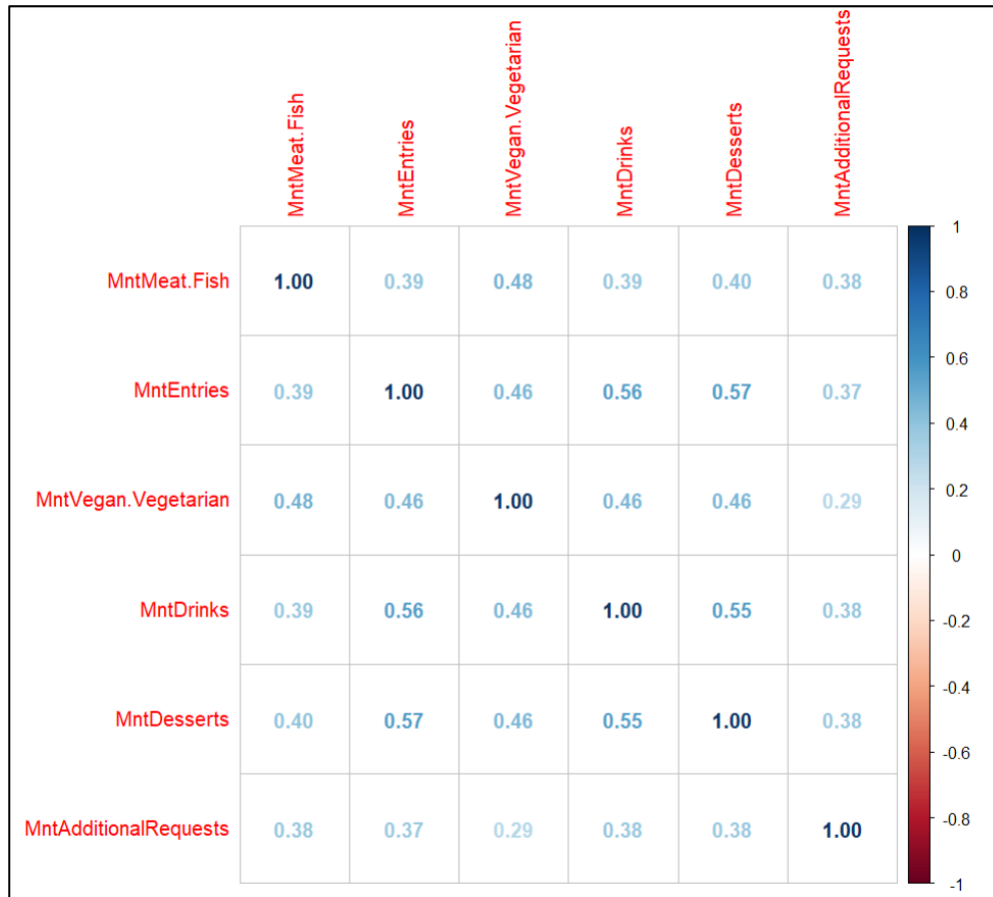
      0      1
0.917 0.083
> prop.table(table(sabor$Response_Cmp4))

      0      1
0.93414286 0.06585714
> prop.table(table(sabor$Response_Cmp5))

      0      1
0.98571429 0.01428571
```

5. Análise exploratória multivariada

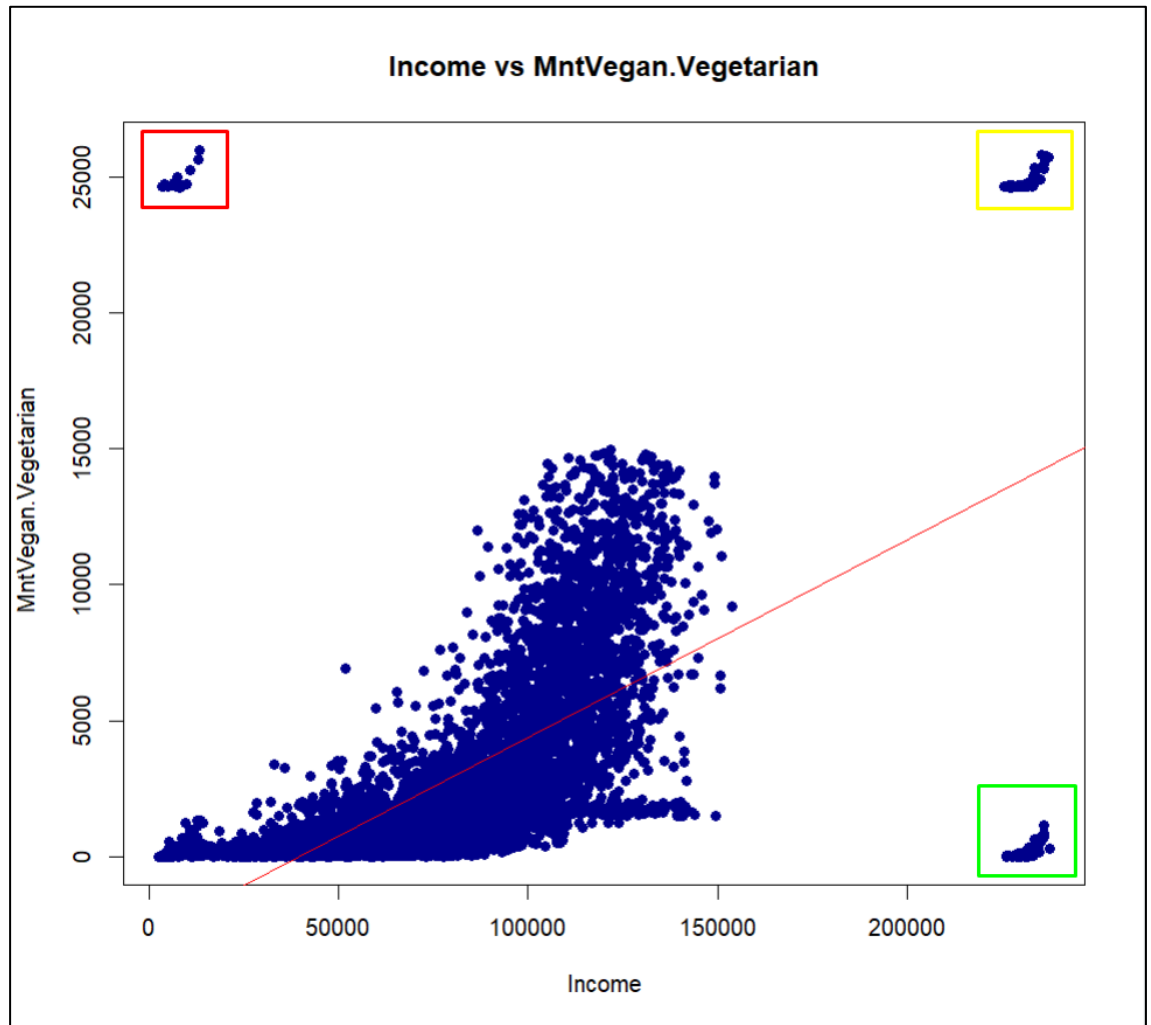
- a. In general, the correlations between the variables are low, indicating that customer's spending on different categories of food and services in the restaurant is not strongly correlated. That is, spending on one category does not automatically predict spending on other categories.



- b. I can see positive trend in the graph, which clearly shows that people with an income of 100k-150k have the highest spendings on vegetarian food. I believe that this is a reasonable data, as meat food is an inexpensive and filling source of healthy nutritional elements. Whereas vegetarian food requires more of it to be consumed to feel filled and it is generally more expensive. Moreover, I can notice three groups of outliers.

The first group is a group of clients whose income is close to zero, however they spent around 25000 on vegetarian food. This indicates either a data error or that clients have been eating at the restaurant for a very long period and have spent this amount of money over that long time.

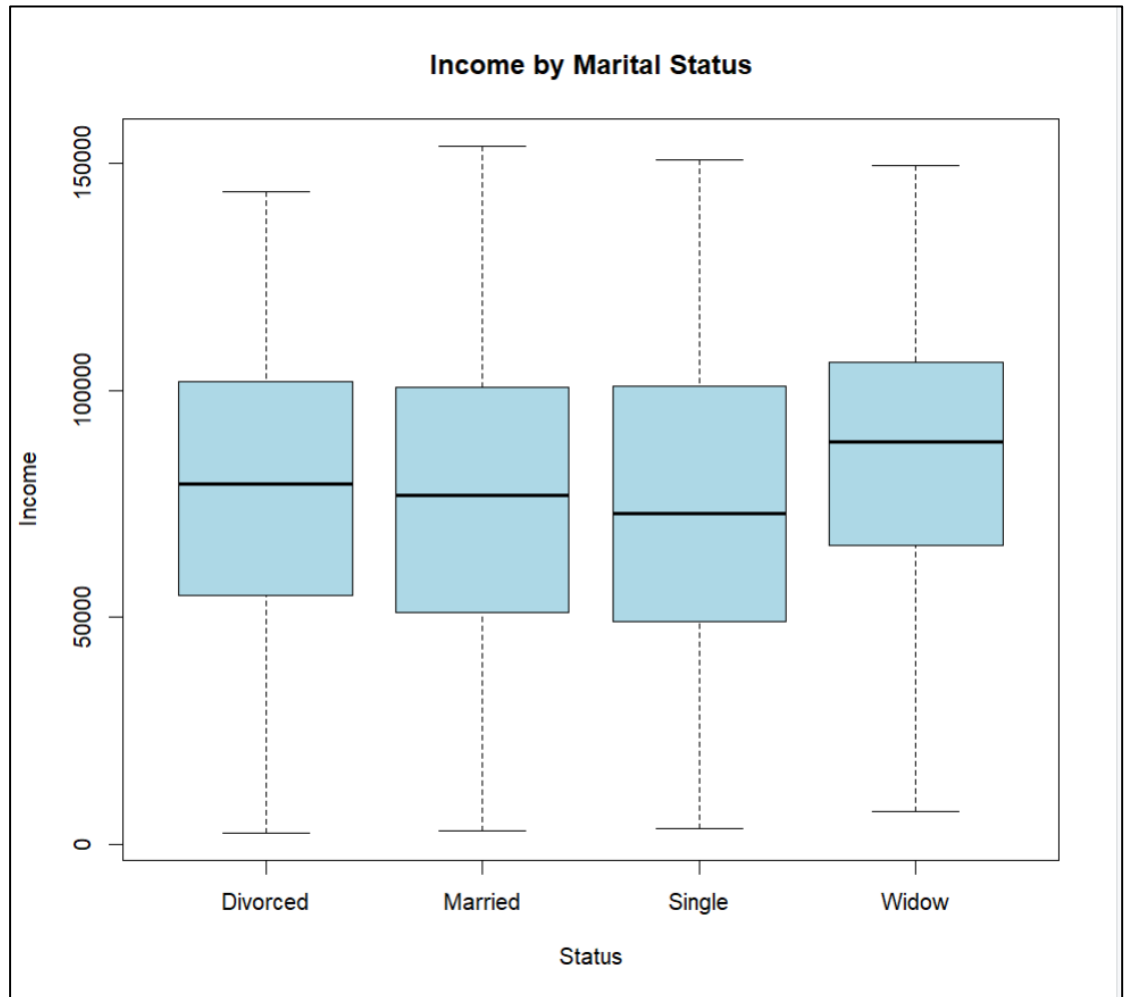
The second group and the third group are the groups of clients whose income is close to maximum. An interesting observation is that these income outliers have no intermediate data on spending on vegetarian food. They either did not eat it almost never, or ate it in quantities close to the maximum for this dataset.



C.

i. In the chart I can observe the approximate equality of income between Divorced, Married, Single groups. However, I can notice some interesting moments:

- Married group has higher maximum number of income over all of the groups.
- Single people have lowest Q1 number and median, however their Q3 and maximum numbers are not too different from others.
- Widow people definitely have higher income overall, with the Q1 and median being much higher than other ones.



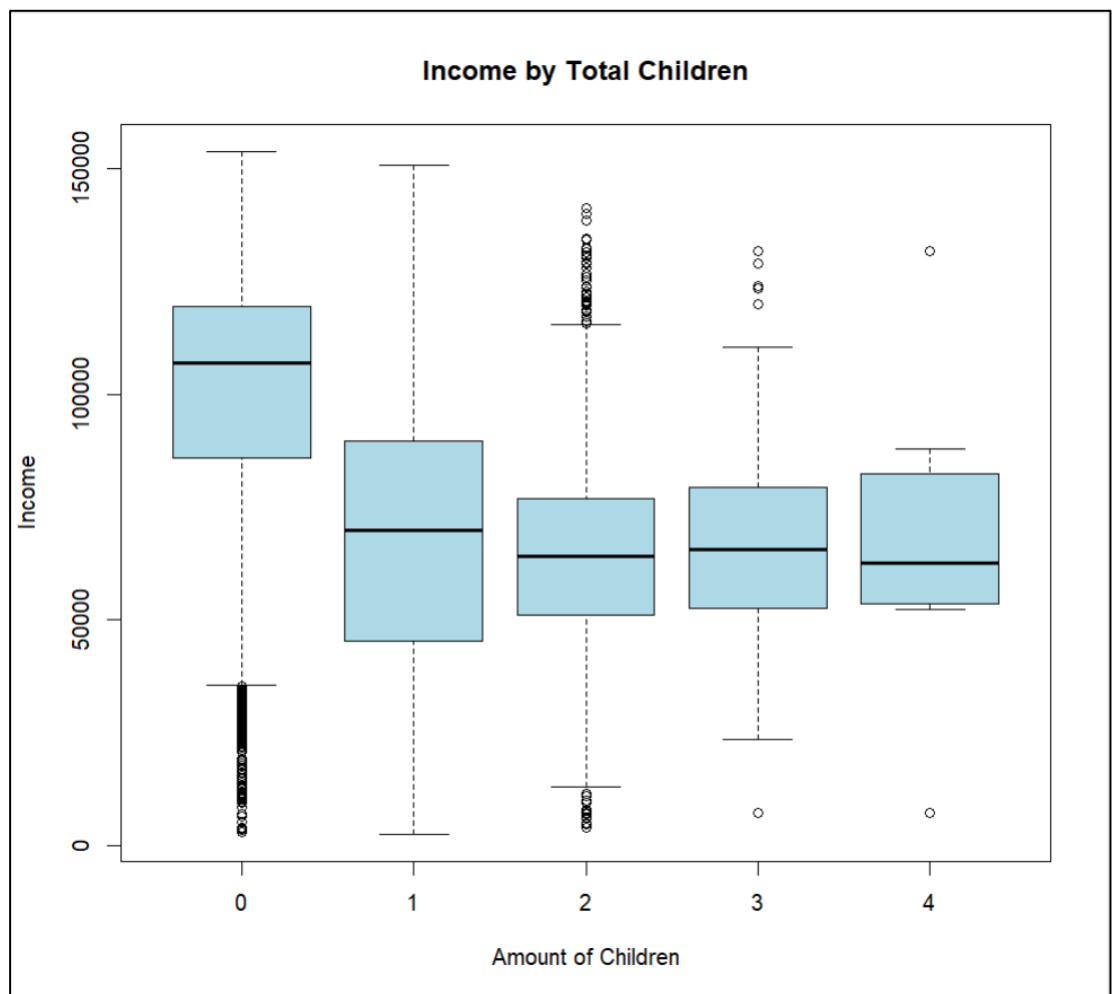
- i. On the chart I can observe very uneven data on which it is difficult to see any global trends. There are many outliers. Let's look by groups:

0 children: Clients with no children have the highest median income, exceeding 100,000. They also have more outliers in the lowest income range, indicating the presence of very low-income clients, despite the high median for this category.

1 child: Data is very symmetrical, with the median being highest among other groups of clients with children.

2 children: The income of this group is lower than the income of clients with 1 child, also I can see that the maximum is much lower, nevertheless there are many outliers present, both at the top and at the bottom. I suppose that in many families, when there are two children, the mother often leaves her job, which can affect the family income.

3 or 4 children: There are fewer outliers in the three- and four-child categories, indicating more homogeneous incomes among these clients. However, clients with four children have a very narrow range of incomes, which is indicated in minimum and maximum being very close to Q1 and Q3.



Análise Inferencial:

1.

- a. H_0 : The mean of amount spent on desserts between clients with and without children is equal. ($\mu_1 = \mu_2$)

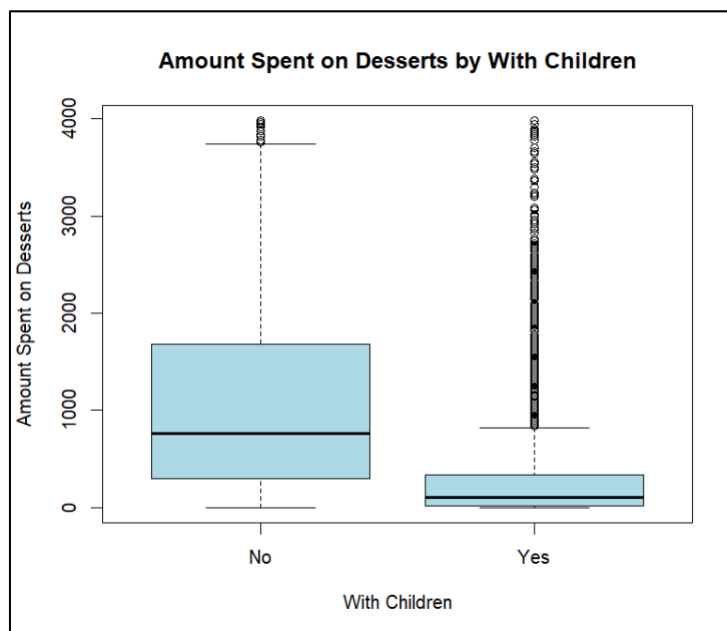
H_1 : The mean of amount spent on desserts between clients with and without children is not equal. ($\mu_1 \neq \mu_2$)

Condições de aplicabilidade e pressupostos:

- ✓ MntDesserts é uma variável quantitativa
- ✓ Existe uma variável categórica (WithChildren) que define 2 grupos independentes (Yes: Com filhos; No: Sem filhos)
- ✓ Dispõe-se de duas amostras de dimensão $n_1 = 2051$, $n_2 = 4863$, ambas superiores a 30, pelo que se pode admitir que a distribuição das médias amostrais é aproximadamente Normal, pelo Teorema do Limite Central

```
data: MntDesserts by WithChildren  
t = 32.751, df = 2595.1, p-value < 2.2e-16
```

Decision: Assuming a significance level of 0.05 ($\alpha = 0.05$), as $p\text{-value} = 2.2e-16 < 0.05$ then **rejecting H_0** . And there is statistical evidence against the hypothesis 0 that the mean of amount spent on desserts between clients with and without children **is not equal**.



The low average spend on desserts among customers with children may indicate that desserts are not a major part of orders for families, perhaps due to budget constraints or preferences for other dishes. The presence of outliers suggests that there is a subgroup of families willing to spend more on desserts. This may be due to special occasions (celebrations, birthdays), or such clients prefer

more expensive desserts.

Restaurant should create dessert offers for families: Given that families spend less on desserts on average, you can offer dessert discounts on main course orders or “dessert for family” packages to motivate them to spend more on desserts.

- b. H_0 : The mean of days from the last visit between clients with and without children is equal. ($\mu_1 = \mu_2$)

H_1 : The mean of days from the last visit between clients with and without children is not equal. ($\mu_1 \neq \mu_2$)

Condições de aplicabilidade e pressupostos:

→ Recency é uma variável quantitativa

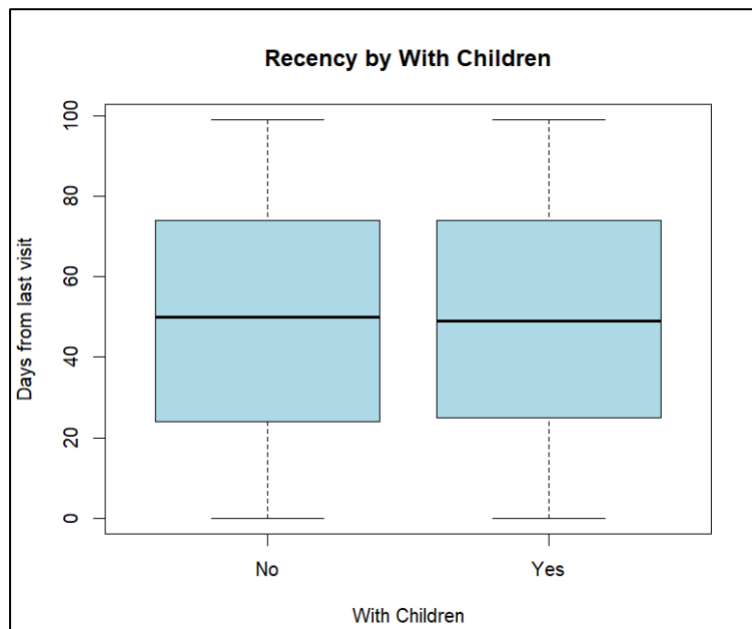
→ Existe uma variável categórica (WithChildren) que define 2 grupos independentes (Yes: Com filhos; No: Sem filhos)

→ Dispõe-se de duas amostras de dimensão $n_1 = 2051$, $n_2 = 4863$, ambas superiores a 30, pelo que se pode admitir que a distribuição das médias amostrais é aproximadamente Normal, pelo Teorema do Limite Central

```
data: Recency by withChildren  
t = 0.40829, df = 3799.8, p-value = 0.6831
```

Decision: Assuming a significance level of 0.05 ($\alpha = 0.05$), as $p\text{-value} = 0.6831 > 0.05$ then **not rejecting H_0** . The mean of days from the last visit between clients with and without children **is equal**.

This means that stability of people visiting the restaurant is not dependent on them having children.



- c. **H₀**: Variables MaritalStatus and WithChildren are independent.
H₁: Variables MaritalStatus and WithChildren are not independent.

```
Pearson's Chi-squared test

data:  table_marital_children
X-squared = 5.6674, df = 3, p-value = 0.129
```

Decision: Assuming a significance level of 0.05 ($\alpha = 0.05$), as p-value = 0.129 > 0.05 then **not rejecting H₀**. Variables MaritalStatus and WithChildren **are independent**.

Although it seems there has to be a dependency, test shows us that there is no. I can see strange data in “Single” category of clients, ~66% of them have children, even if it is not possible, because we have categories “Divorced” and “Widow” for people who have a child, but don’t have a partner.

	No	Yes
Divorced	176	455
Married	1317	3202
Single	476	1048
widow	82	158

2. Esta afirmação não é verdadeira.

H₀: Mean of the amount spent on Meat and Fish and mean of the amount spent on Vegan and Vegetarian are equal. ($\mu_1 = \mu_2$)

H₁: Mean of the amount spent on Meat and Fish and mean of the amount spent on Vegan and Vegetarian are not equal. ($\mu_1 \neq \mu_2$)

```
welch Two Sample t-test

data:  sabor_upd$MntMeat.Fish and sabor_upd$MntVegan.Vegetarian
t = 7.969, df = 13813, p-value = 1.725e-15
```

Assuming a significance level of 0.05 ($\alpha = 0.05$), as p-value = 1.725e < 0.05 then **rejecting H₀**. And there is statistical evidence against the hypothesis 0 that the mean of the amount spent on Meat and Fish and the mean of the amount spent on Vegan and Vegetarian **are not equal**.

3. H_0 : The mean of purchases with promotion offers is equal to 1. ($\mu_1 = 1$)
 H_1 : The mean of purchases with promotion offers is more than 1. ($\mu_1 > 1$)

Condições de aplicabilidade e pressupostos:

- ✓ NumOfferPurchases é uma variável quantitativa
- ✓ Dimensão da amostra, $n = 6914 > 30$, pelo que a distribuição das médias amostrais pode ser considerada aproximadamente normal, pelo Teorema do Limite Central

One Sample t-test

```
data: sabor_upd$NumOfferPurchases  
t = 54.549, df = 6913, p-value < 2.2e-16
```

Assuming a significance level of 0.05 ($\alpha = 0.05$), as $p\text{-value} = 2.2e-16 < 0.05$ then

rejecting H_0 . And there is statistical evidence against the hypothesis 0 that the mean of purchases with promotion offers **is more than 1**.

From this result I can conclude that, on average, customers made at least one purchase using the offers, they know about offers and they use them. So, I can suggest that it is necessary to continue utilizing and developing them in order to achieve better numbers on the use of offers.

4. I would like to hypothesize that overall spendings in the restaurant **is not the same** between clients with and without children. To achieve that, I created a variable MntTotal.

H₀: The mean of MntTotal between clients with and without children is equal. ($\mu_1 = \mu_2$)

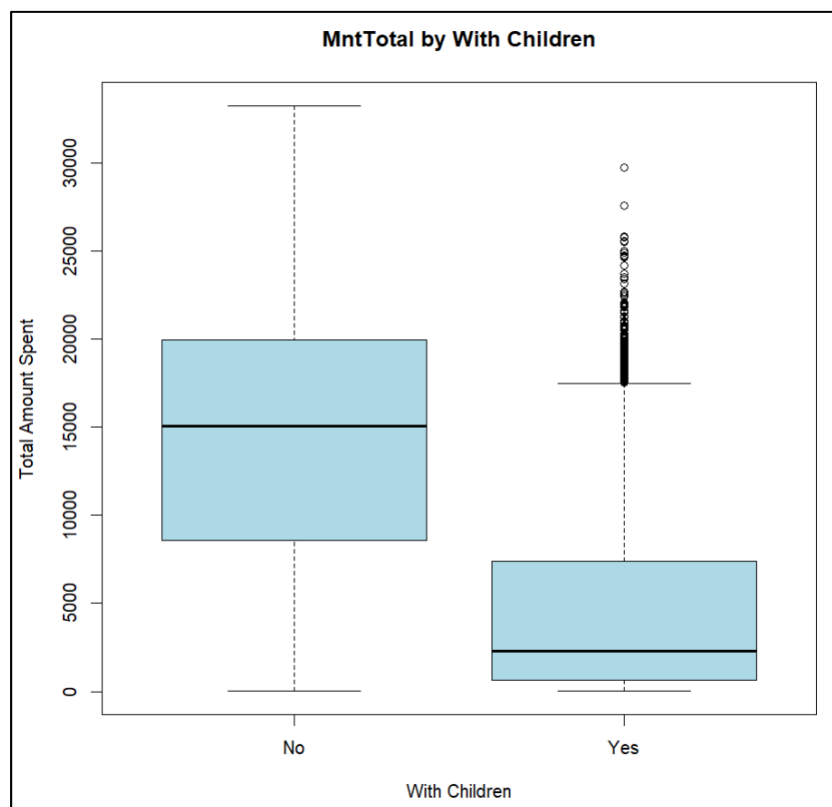
H₁: The mean of MntTotal between clients with and without children is not equal. ($\mu_1 \neq \mu_2$)

```
Welch Two Sample t-test  
  
data: MntTotal by WithChildren  
t = 50.261, df = 2811.2, p-value < 2.2e-16
```

Assuming a significance level of 0.05 ($\alpha = 0.05$), as $p\text{-value} = 2.2e-16 < 0.05$ then **rejecting H₀**. And there is statistical evidence against the hypothesis 0 that the mean of MntTotal between clients with and without children **is not equal**.

I've confirmed my hypothesis, but in order to make suggestions regarding our scenario I have to create a boxplot chart, to see where the difference is going.

I can see that the amount spent by clients with children is **significantly lower**.



I suggest creating offers for families. For example, cheap lunch combo, that will include a lunch for parents and children menu for kids. Therefore, I can increase the amount spent by family clients.

5.

a. **One-Way ANOVA ou Análise de Variância Simples:**

b. ☒ A variável *MntAdditionalRequests* é quantitativa

☒ Existe uma variável categórica *Marital_Status* que define os k grupos (populações) independentes

☒ **Pressupostos:**

☒ As amostras provenientes dos quatro grupos populacionais **são independentes**. Os grupos são mutuamente exclusivos;

☒ Normalidade da distribuição nos quatro grupos:

H₀: *MntAdditionalRequests* **follows** a normal distribution in each of marital status groups.

H₁: *MntAdditionalRequests* **does not follow** a normal distribution in each of marital status groups.

Assuming a significance level of 0.05 ($\alpha = 0.05$), as p-value = $2.2e-16 < 0.05$ then **rejecting H₀**. And there is statistical evidence against the hypothesis 0 that *MntAdditionalRequests* **does not follow** a normal distribution in each of marital status groups.

It is not possible to use ANOVA in this case.

```
$Divorced
      Shapiro-Wilk normality test
data:  x[[i]]
W = 0.78998, p-value < 2.2e-16

$Married
      Shapiro-Wilk normality test
data:  x[[i]]
W = 0.77741, p-value < 2.2e-16

$Single
      Shapiro-Wilk normality test
data:  x[[i]]
W = 0.75921, p-value < 2.2e-16

$Widow
      Shapiro-Wilk normality test
data:  x[[i]]
W = 0.81009, p-value < 2.2e-16
```

a. I will use an alternative test of **Kruskal-Wallis**.

b. ☒ **Pressupostos:**

☒ As amostras provenientes dos quatro grupos populacionais **são independentes**. Os grupos são mutuamente exclusivos;

H₀: The mean of *MntAdditionalRequests* **is equal** across population groups.

H₁: The mean of *MntAdditionalRequests* **is not equal** across population groups.

```
data:  MntAdditionalRequests by Marital_Status
Kruskal-Wallis chi-squared = 3.8736, df = 3, p-value = 0.2754
```

Decision: Assuming a significance level of 0.05 ($\alpha = 0.05$), as p-value = $0.2754 > 0.05$ then **not rejecting H₀**.

The mean of *MntAdditionalRequests* **is equal** across population groups.

Based on the results of the test, I can assume that the amount that people spend on additional requests does not depend on their marital status.

- c. The lack of statistically significant differences may indicate that marital status has no impact on the amount spent on additional orders. This may mean that customers, regardless of their marital status, have similar consumption habits regarding additional orders.

ERRO TIPO II = Não rejeitar H_0 quando H_0 é falsa (Não Rejeitar H_0 | H_0 falsa)